

Textbook Questions

Q1: Explain the importance of reproduction in organisms.

Ans: Reproduction is an essential process which is essential for existence and continuity of life and species, generation after generation.

Q2: Describe the process of fertilisation in human beings.

Ans: In the process of fertilisation, sperms come in contact with an ova (egg), one of the sperms may fuse with the egg. The nuclei of the sperm and the egg fuse to form a single nucleus resulting in the formation of a fertilised egg called zygote..

Q5: Give two differences between a zygote and a foetus.

Ans:

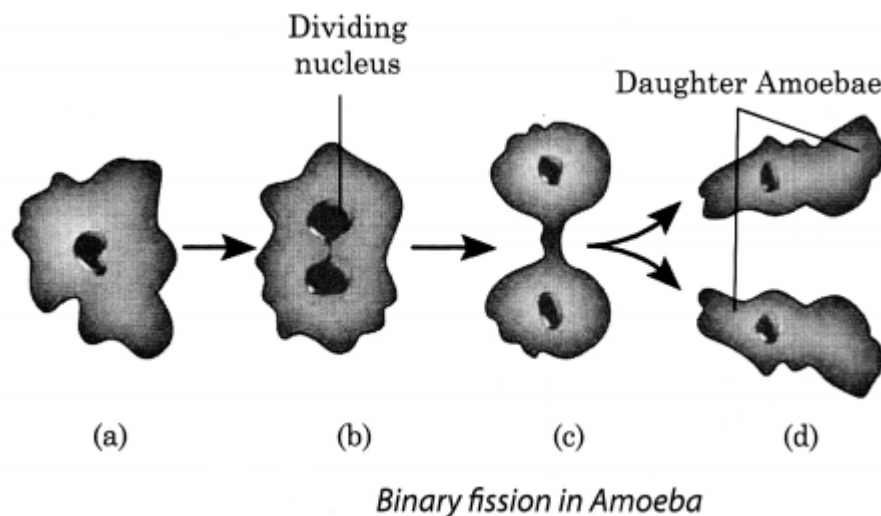
Zygote	Foetus
(i) It is a single celled, i.e., it contains only one cell.	(i) It is multicelled, i.e., it contains many cells.
(ii) It is formed by the fusion of male gamete (sperm) and female gamete (ova/egg).	(ii) It is formed by the repeated division of the zygote.

Q6: Define asexual reproduction. Describe two methods of asexual reproduction in animals.

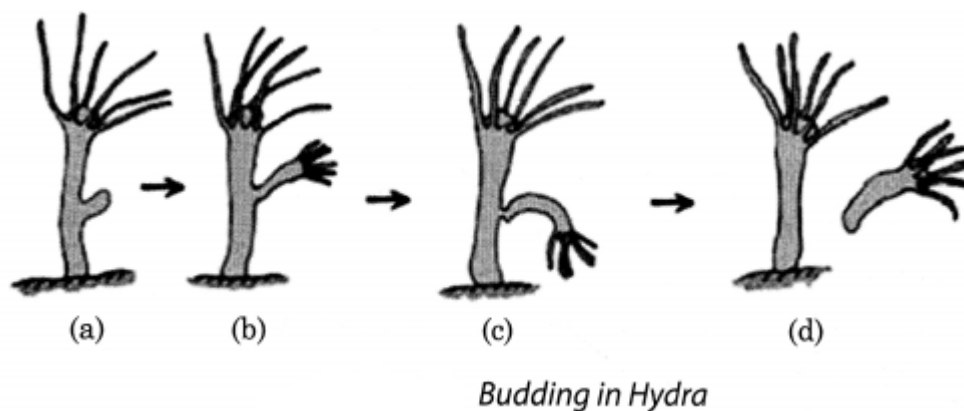
Ans: The mode of reproduction in which only a single parent is involved is called asexual reproduction. In this type of reproduction, sex cells (gametes) are not produced and no fertilisation takes place. Asexual reproduction takes place in Amoeba, Hydra, yeast, starfish, sponges, etc.

The two methods of asexual reproduction are:

(i) Binary fission: In binary fission, a single parent cell is divided into two equal individual cells e.g. in Amoeba. The nucleus of the parent cell divides into two nuclei, followed by the division of its body into two, each part receiving one nucleus and develops into separate individual.



(ii) Budding: In budding, the organism develops a bulge called bud which further develops into an adult organism and separates itself from the parent body to lead an independent life. This type of reproduction is shown in Hydra etc



Q7: In which female reproductive organ does the embryo get embedded?

Ans: Uterus is the female reproductive organ where the embryo gets embedded.

Q8: What is metamorphosis? Give examples.

Ans: The transformation of larva into an adult involving sudden and abrupt changes in the body of an animal during the life cycle is called metamorphosis. E.g., frog and butterfly.

Q9: Differentiate between internal fertilisation and external fertilisation.

Internal Fertilisation	External Fertilisation
(i) The fusion of male gamete (sperm) and female gamete (ova) occurs inside the body of a female, e.g. in human beings, birds, and mammal.	(i) The fusion of male gamete and female gamete takes place outside the body of a female, e.g. in frog, fish and starfish.

(ii) The female partner lays either fertilised eggs or a fully grown young one.	(ii) The female partner discharges unfertilised eggs.
(iii) Fewer gametes are produced, but chances of fertilization are high.	(iii) Large number of gametes are produced to compensate for loss, but chances of fertilization are low.

